

CS 571: Programming Practicum

Course Code	CS 571
Course Name	Programming Practicum
L-T-P-C	1-0-3-3
Prerequisites	Programming experience in any language (C/C++/Matlab/Python etc.)
Intended for	MTech./M.S./Ph.D.
Distribution	Core for M.Tech. CSP, Elective for M.S./Ph.D.
Approval	22nd Senate (Jun 2019)

Course Contents

1 lecture per week, followed by 3 hours of lab.

- **Review of general programming constructs:** Loops, conditionals, recursion, file i/o, data structures: strings, tuples, lists, dictionaries. [4 Lectures]
- **Introduction to scientific computing:** Numerical precision in programs, IEEE 754 floating point representation, introduction to NumPy and Scipy. [1 Lecture]
- **Data manipulation:** Pandas, handling large data files. [1 Lecture]
- **Data visualization:** Various types of plots: histograms, scatter plots, box plots etc. Datasets can be provided and plots can be created from them. [1 Lecture]
- **Object oriented programming:** Classes and objects, inheritance. [1 Lecture]
- **Implementing well known programs:** Matrix factorizations, solving large order linear systems of equations, least squares approximations, simulating binary channels, signal denoising, K-means clustering, classification using Bayes rule etc. [4 Lectures]
- **Introduction to parallelization:** Cuda programming (conceptual level only). [1 Lecture]
- **Program analysis:** Performance tuning, profiling of programs, identifying performance bottlenecks. [1 Lecture]

Text Books

1. Jake Vanderplas, **Python Data Science Handbook**, O Reilly, 2017

References

1. Michael Dawson, **Python Programming for the Absolute Beginner**, Third Edition
2. Allen Downey, Jeffrey Elkner, Chris Meyers, **How to Think Like a Computer Scientist: Learning with Python**, Green Tea Press, 2016
3. Muller and Guido, **Introduction to machine learning with Python**, O Reilly, 2017